Question Paper

Exam Date & Time: 27-Dec-2025 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026
SUBJECT: CSS 2103/CSS_2103 DATA ANALYTICS
(SPL: COMPUTER SCIENCE AND ENGINEERING / ARTIFICIAL INTELLIGENCE / CYBER SECURITY / DATA SCIENCE / INFORMATION TECHNOLOGY)

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A) (a) A mobile accessories retailer analyses six recent online orders to understand attachment behaviour around new smartphone purchases. The transaction log are shown in Table Q.1A (5)
The product codes map to real items as follows:
a = Smartphone
b = Protective Case
c = Screen Protector
d = Power Bank
e = USB Cable

Table Q.1A

Transaction ID	Items
T1	a, b, d, e
T2	b, c, e
T3	a, b, d, e
T4	a, b, c, e
T5	a, b, c, d, e
T6	b, c, d

(b) Derive all possible frequent k-itemsets with minimum support count=3 and following the Apriori property, list pruned itemsets at each stage.
(c) Generate a minimum of 5 possible association rules using the longest frequent k-itemsets and verify whether the generated rules are strong with the confidence threshold=60%.

Course Outcome: "CO4" / Cognitive Level: "Application"

1B) A hospital laboratory claims that a new COVID testing kit correctly identifies infection 95% of the time.A researcher believes that the true accuracy is less than 95% and decides to test this claim at a 5% significance level. A random sample of n = 50 test results gives a sample mean accuracy of 0.93. The population standard deviation is known to be 0.04. At the 5% significance level, test whether the hospital laboratory's claim is justified. (critical value $Z_{0.05} = -1.645$). (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

1C) Consider the scenario of mobile phone sales data, the mobile phone retailer analyzed monthly sales (2)

data over the period of 3 years. The data showed consistent upward movement due to market growth, with a little drop due to global electronic chip shortages. Identify the component of time series that represents the market growth, and name the time series component denote temporary drop?

Course Outcome: "CO5" / Cognitive Level: "Understand"

2A) Per acre production data (in tons) for three varieties of wheat grown using four different fertilizers, each combination replicated twice, are given in Table Q. 2A. Test whether the variety differences and fertilizer differences are significant at 5% level. (5)

Table Q. 2A.

	Per Acre Production (in Tons)			
	Fertilizer 1	Fertilizer 2	Fertilizer 3	Fertilizer 4
Variety 1	8	9	6	10
	9	8	7	11
Variety 2	7	8	5	9
	6	9	4	10
Variety 3	10	11	8	12
	9	12	7	13

Course Outcome: "CO3"

2B) Suppose you have $L_1 = \{I_1, I_2, I_3, I_4, I_5, I_6\}$ and the following transactions: (3)

T_1: $\{I_1, I_2, I_3, I_6\}$
T_2: $\{I_2, I_4, I_5\}$
T_3: $\{I_1, I_3, I_4, I_6\}$
T_4: $\{I_2, I_3, I_5, I_6\}$
T_5: $\{I_1, I_2, I_4, I_5\}$.

Use the hash function

$$h(x,y) = ((order(x) \times 11) + order(y)) \bmod 10$$

(a) Construct the hash table for all 2-itemsets.
(b) Determine bucket counts.
(c) If **min support = 4**, specify which buckets are pruned and list the remaining candidate itemsets C_2.

Course Outcome: "CO4"

2C) Differentiate between Mean Absolute Error (MAE) and Mean Squared Error (MSE). Assume two model with given MAE and RMSE . Interpret the results and identify the better forecasting model. Justify your answer briefly. (2)

• Model A: MAE = 5.2, RMSE = 6.8
• Model B: MAE = 4.9, RMSE = 5.0

Course Outcome: "CO5" / Cognitive Level: "Application"

3A) i) Find the elements of box plot using the following data:78, 91, 68, 54, 82, 47, 60, 78, 97, 58, 62, 50, 50, 63, 92, 75 79, 63, 42, 61. (5)
ii) Why is it sometimes necessary to create subsets from a large dataset before performing analysis?
iii) Measure the strength of the relationship between the ranks assigned by two different interviewers as shown below.



Candidate	Interviewer 1	Interviewer 2
A	1	1
B	2	2
C	3	4
D	4	3
E	5	6
F	6	5
G	7	8
H	8	7
I	9	10
J	10	9
K	11	12
L	12	11

Course Outcome: "CO1" / Cognitive Level: "Analysis"

3B) A supermarket dataset has the following transactions: (3)
T1 = {Milk, Bread, Butter}
T2 = {Milk, Bread}
T3 = {Milk, Butter}
T4 = {Bread, Butter}
T5 = {Milk, Bread, Butter}
i) Compute the support for each of the following:
(a) {Milk, Bread}
(b) {Milk, Butter}
(c) {Bread, Butter}
ii) Identify which itemsets are frequent if the minimum support threshold = 60%.
iii) Determine which among them are maximal frequent itemsets.

Course Outcome: "CO4"

3C) The sales of a retail store for five consecutive weeks are as follows: 100, 110, 95, 120, 130. Compute the 3-week moving average forecast for week 6. Interpret the whether this method would be appropriate if a strong upward trend continues. (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

4A) Given the following transaction database: (5)

TID	Items
1	A, B, D
2	B, C, E
3	A, B, C, E
4	B, E

(a) Using the FP-Growth algorithm, mine all frequent itemsets with a minimum support of 2.
• Construct the FP-tree
• Show the conditional pattern bases and conditional FP-trees
• List all frequent itemsets
(b) State the advantages of the pattern growth approach over the Apriori algorithm.

Course Outcome: "CO4" / Cognitive Level: "Application"

4B) A beverage company tracks **monthly sales (in units)** of a new energy drink: (3)

Month	Sales
1	80

2	95
3	100
4	110
5	120

Using **Single Exponential Smoothing (SES)** with a smoothing constant **α = 0.4**, and assuming the forecast for **Month 1 = 80 units,**

(a) Calculate the forecast for Months 2 to 6.

(b) Compute the **Mean Absolute Error (MAE)** for the forecasts from **Months 2 to 5.**

Course Outcome: "CO5" / Cognitive Level: "Application"

4C) A popular online bookstore analysed 800 customer purchase transactions and obtained the following data: (2)

Itemset	Number of Transactions
{Fiction Book}	360
{E-Reader}	300
{Fiction Book, E-Reader}	180

(a) Calculate the pattern evaluation measures Lift and Kulc for the association rule:
Fiction Book → E-Reader

(b) Interpret what the obtained Lift and Kulc values indicate about the relationship between Fiction Book and E-Reader purchases.

Course Outcome: "CO4" / Cognitive Level: "Application"

5A) A manufacturing unit tracks its quarterly production output. The data for the first four quarters are given below: (5)

Period (t)	Actual Production (Yt)
Q1	100
Q2	120
Q3	130
Q4	150

Use smoothing constants =0.2 and =0.4. Applying Holt's Method

(a) Calculate the initial Level (L1) and Trend (T2).

(b) Calculate the Level (L2) and Trend (T3) for periods 2 and 3, respectively.

(c) Calculate the Forecast for period 4 (F4).

(d) Differentiate between Double and triple exponential smoothing techniques.

Course Outcome: "CO5" / Cognitive Level: "Application"

5B) A transaction database is given as follows: (3)

Transaction ID	Items Purchased
T1	A, B, C
T2	A, C, D
T3	B, C, E
T4	A, B, C, E
T5	A, B, D, E

(a) List all **frequent itemsets** having minimum support = 2.

(b) Identify the **closed frequent itemsets** and **maximal frequent itemsets** from the result.

(c) Explain why mining **closed** and **max patterns** is more efficient than mining all frequent itemsets.

Course Outcome: "CO4"

5C) A data scientist is analysing the following cholesterol level (mg/dL) readings of 12 patients: [145, 155, 160, 162, 165, 172, 180, 190, 200, 220, 240, 260]. Apply equal-width and equal-frequency binning with 3 bins. Using histogram analysis, explain which binning approach (width-based or frequency-based) better represents the data distribution and why. (2)

Course Outcome: "CO3"

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Event Name	Subject
MAKE-UP EXAM-ODD SEMESTER 2025 MIT MANIPAL	CSS 2103:DATA ANALYTICS

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	3	CO 3	10
3	4	CO 4	21
4	5	CO 5	14
		Total	**50**

* NA - CO Not Selected

Event Name	Subject
MAKE-UP EXAM-ODD SEMESTER 2025 MIT MANIPAL	CSS 2103:DATA ANALYTICS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	1	2(5.88%)
3	Application	7	24(70.59%)
4	Analysis	2	8(23.53%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	34 (100%)